

Production-Grade Bill of Materials (BOM) & 3-Year Lifecycle Cost Analysis

Modernized Philippine Government Electronic Procurement System (mPhilGEPS) Phase 2

Target Cloud Platform: Google Cloud Platform, Gemini Enterprise & Google Sovereign AI Stack
Primary Region: Singapore (asia-southeast1) | **Disaster Recovery Region:** Jakarta (asia-southeast2)
Statutory Governance: RA 12009 (New Government Procurement Act / NGPA) & RA 10173 (Data Privacy Act)
Currency Benchmark: USD (\$) and Philippine Peso (PHP ₱) at official budgetary baseline rate: **\$1.00 USD = ₱65.00 PHP**

1. Executive

Strategic Architectural Value

1. **Zero External AI Egress Fees:** By utilizing **BigQuery ML (ARIMA_PLUS)** and **Cloud Spanner Graph** natively in-database, predictive demand forecasting and cartel detection execute directly where the sovereign data resides. No expensive data movement to third-party AI APIs is incurred.
2. **Serverless & Autopilot Elasticity:** Core microservices run on **GKE Autopilot** and **Cloud Run**, automatically scaling down during non-working hours and scaling up to handle peak traffic surges of **5,500+ concurrent sessions** without over-provisioning static VMs.
3. **Mandated 10-Year WORM Compliance:** Archival records are held in **Cloud Storage Archive Tier with Bucket Lock in Compliance Mode** at only **\$0.0024/GB/month**, fulfilling Commission on Audit (COA) and RA 12009 legal retention rules at negligible storage cost.
4. **Dual-Channel Frontend Deployment:</**

3. Categorized Production Bill of Materials (Singapore: asia-southeast1)

Category 1: Core Compute, Container Platform & Confidential Enclaves

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
1.1	GKE Autopilot (vCPU)	55 vCPU monthly average across 10 domain microservices (HPA autoscaling up to 80 vCPU)	\$0.0535 / vCPU-hr	\$2,148.03	₱139,621.95	\$25,776.36
1.2	GKE Autopilot (Memory)	150 GiB RAM monthly average across pods				

Category 2: Relational, Graph & Distributed In-Memory Databases

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
2.1	Cloud Spanner (Compute)	1,200 Processing Units (1.2 Nodes equivalent) in asia-southeast1 . Houses contract ledger & Spanner Graph	\$0.0009 / PU-hr (\$0.90/1000 PU-hr)	\$788.40	₱51,246.00	\$9,460.80
2.2	Cloud Spanner (Storage)	500 GB Regional SSD Storage	\$0.30 / GB-mo	\$150.00</		

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
		holds & ZSET auction ticker				
SUB	Category 2 Subtotal	Relational, Graph & Distributed Databases	—	\$3,102.80	₱201,682.00	\$37,233.60

Category 3: Enterprise Data Warehouse, In-Database ML & BI

Item #	GCP Component / SKU	Technical Sizing &
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Note on BI Alternatives: If PS-DBM elects to deploy **Looker Studio Pro** with 30 named creator licenses (\$9/user/mo = \$270.00/mo) in lieu of Looker Core, Category 3 subtotal drops to **\$1,002.33 / month** (₱65,151.45 PHP), reducing the total monthly bill by \$4,730.00 USD.

Category 4: Sovereign AI & Multi-Agent Ecosystem

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
4.1	DocAI Custom Extractor	7,500 pages/mo (SEC GIS, DTI, PCAB Construction Licenses) for GOP-OMR accreditation	\$50.00 / 1,000 pages	\$375.00	₱24,375.00	\$4,500.00
4.2	DocAI Form Parser	2,500 pages/mo (BIR Tax Clearance Certificates with barcode validation)	\$30.00 / 1,000 pages	\$75.00	₱4,875.00	

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
SUB	Category 4 Subtotal	Sovereign AI & Multi-Agent Ecosystem	—	\$702.58	₱45,667.70	\$8,430.96

Category 5: Edge Perimeter, Networking & API Management

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
5.1	Apigee API Management	1 Production Environment (\$0.20/hr) + 4,000,000 API calls/mo for external integrations (BTMS, SEC, BIR, Banks)	Hourly env + Tiered API call fee	\$796.00	₱51,740.00	\$9,552.00
5.2	Cloud Armor Enterprise	Managed Protection Plus: 24/7 DDoS mitigation, OWASP Top 10 rules, rate limiting, and DDoS billing protection	\$3,000.00 / mo	\$		

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
5.7	Internet Egress (Premium)	5 TB / month un-cached outbound Internet traffic (responses to bidders, API payloads, document downloads)	\$0.120 / GB	\$600.00	₱39,000.00	\$7,200.00
SUB	Category 5 Subtotal	Edge Perimeter, Networking & API Management	—	\$4,872.10	₱316,686.50	\$58,465.20

Category 6: Sovereign Cryptographic Root of Trust &

Category 7: Cloud Storage & 10-Year WORM Legal Compliance

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
7.1	Cloud Storage (Standard)	10,000 GB (10 TB) active storage for vendor dossiers, APP-CSE workbooks, and Virtual Store catalog media	\$0.023 / GB-mo	\$230.00	₱15,164.50	\$2,760.00
7.2	Standard Storage Operations	500,000 Class A operations (write/list) + 2,000,000 Class B operations (read)	Class A (\$0.05/10k) + Class B (\$0.004/10k)			

Category 8: Enterprise Observability, Governance & 24/7 SLA Support

Item #	GCP Component / SKU	Technical Sizing & Configuration	Unit Price (USD)	Monthly (USD)	Monthly (PHP @ ₱65)	Annual (USD)
8.1	Cloud Logging Ingestion	250 GB application & audit logs/mo (first 50 GB free, 200 GB billed) with 30-day retention	\$0.500 / GiB	\$100.00	₱6,500.00	\$1,200.00
8.2	Cloud Monitoring Metrics	500M non-free metric samples/mo for GKE microservices, HPA autoscale, and alert policies	\$0.258 / 1M samples			

5. Cost Optimization & Committed Use Discounts (CUDs)

By committing to a 1-Year or 3-Year baseline commitment for predictable production workloads (GKE Autopilot Compute, Cloud Spanner Processing Units, and AlloyDB instances), PS-DBM can capture major budget savings:

Discount Optimization Table

Workload Component	On-Demand Monthly	1-Year CUD Savings (~30%)	1-Year CUD Net	3-Year CUD Savings (~50%)	3-Year CUD Net
GKE Autopilot Compute	\$2,788.39	-\$836.52	\$1,951.87	-\$1,394.20	\$1,394.19
Cloud Spanner (1,200					

7. Statutory Value Realization & ROI (RA 12009 Compliance)

Investing in Google Cloud's modern AI and data architecture delivers concrete operational savings and risk reductions for PS-DBM:

1. Elimination of Legacy Relational Database Licensing:

- Transitioning away from legacy proprietary database licenses (e.g., Oracle EE / Microsoft SQL Server) to **AlloyDB Enterprise** and **Cloud Spanner** yields an estimated **₱20,000,000 to ₱28,000,000 in avoided multi-year software licensing and core maintenance fees.**

2. 85% Reduction in Manual Dossier Verification Labor: